

⚡ WD002 Week 5 — English Worksheet

Topic: Conditionals — Checking Answers · **Course:** Web Dev Basics: JavaScript · **Time:** about 45 minutes

This week your code **decides**. `if / else` runs different code depending on a condition. `===` checks if two things are exactly equal. `&&` (AND) needs *both* sides true; `||` (OR) needs *either* side. You will build a number-guessing game that tells the user "higher" or "lower".

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
<script>
  let guess = 7;
  if (guess === 7) {
    console.log("Correct!");
  } else {
    console.log("Wrong");
  }
</script>
```

Which message prints?


```
<script>
  let guess = 4;
  let secret = 10;
  if (guess < secret) {
    console.log("Try bigger");
  } else {
    console.log("Try smaller");
  }
</script>
```

The guess is smaller than the secret. Which message prints?


```
<script>
  let answer = 5;
  let name = "Suho";
  if (answer === 5 && name === "Suho") {
    console.log("Both correct!");
  }
</script>
```

&& needs both sides true. Does the message print? What if the name were different?


```
<script>
  let day = "Saturday";
  if (day === "Saturday" || day === "Sunday") {
    console.log("Weekend!");
  }
</script>
```

|| needs only one side true. Does the message print? What about **Monday** ?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should celebrate only when the guess is exactly 7. Right now it celebrates **every** time, even with the wrong number.

```
<script>
  let guess = 3;
  if (guess = 7) {
    console.log("Correct!");
  }
</script>
```

Hint: one equals sign *sets* the variable to 7; you wanted to *compare*. Use `===`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should say `Try bigger` when the guess is too low and `Try smaller` when it is too high. Right now it says `Try smaller` even when the guess is too low.

```
<script>
  let guess = 2;
  let secret = 8;
  if (guess > secret) {
    console.log("Try bigger");
  } else {
    console.log("Try smaller");
  }
</script>
```

Hint: look closely at the  sign — it is pointing the wrong way for "too low".

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A special prize should appear only when the answer is right **and** the player typed their name. Right now the prize shows when *either* one is true.

```
<script>
  let answer = 5;
  let name = "";
  if (answer === 5 || name === "Mina") {
    console.log("Prize unlocked!");
  }
</script>
```

Hint: "and" is not the same as "or". Which operator means both must be true?

Write the fixed code:

Why was it wrong? Why does your fix work?

1 and 100.

- Tell them `Correct!` , `Try a bigger number` , or `Try a smaller number` using `if / else if / else` .

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First, I picked a secret number and ...

My `if / else` checks ...

To count the tries, I ...

The game tells the user "bigger" or "smaller" by ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) while your game runs. Talk through it like you are teaching someone who has never seen it. Try to use these words: **if**, **else**, **condition**, **compare**, **guess**.

1. Make a guess that is too low and show the message.
2. Make a guess that is too high and show the message.
3. Win the game and show the try count.
4. Read one condition out loud and say what makes it true.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.